

Gaussian Mixture Models: From First Principles

Gaussian Mixture Models, or GMMs, are probabilistic clustering models. Instead of assigning each point only to one cluster with full certainty, GMM gives probabilities.

Why GMM Is Needed

K-Means creates hard clusters and assumes roughly circular groups. GMM can model elliptical clusters and uncertainty.

Gaussian Components

Each cluster is represented by a Gaussian distribution with its own mean and covariance.

Soft Assignment

A point can belong 70% to one cluster and 30% to another. This is useful when clusters overlap.

Core Formula

A mixture model combines several Gaussian distributions:

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)$$

Here, π_k is the weight of component k , μ_k is its mean, and Σ_k is its covariance.

Expectation-Maximization

GMMs are usually trained with EM. The expectation step estimates cluster probabilities. The maximization step updates Gaussian parameters.

When to Use It

Use GMM when clusters overlap or when probabilistic cluster membership is useful. Avoid it when the Gaussian assumption is not reasonable.